THE COMPLETE UNIVERSAL AI PROMPT ENGINEERING GUIDELINES

A concise, practical handbook for GPT-5.6 Luna, Gemini 3.8 Flash and other advanced AI models

Purpose

To improve the accuracy, relevance, completeness, efficiency, creativity and reliability of AI-generated results across research, learning, programming, analysis, business, writing and multimodal tasks.

Core principle

Give the AI the information that materially affects the result: a clear objective, relevant context, mandatory requirements, constraints, an appropriate execution approach, verification criteria and the required output.

This handbook condenses the original 33-page guideline. Repeated explanations and duplicate examples have been removed while retaining the core principles, model-specific guidance, compulsory double-verification protocol and reusable master prompt.

1. The Fundamentals of Effective Prompting

The five essentials

1. Context: What does the AI need to know to understand the situation?

2. Objective: What exactly must it accomplish?

3. Instructions: How should it approach the task?

4. Constraints: What must it follow, preserve or avoid?

5. Success criteria: How will the result be judged?

Universal prompting formula

ROLE + CONTEXT + OBJECTIVE + REQUIREMENTS + CONSTRAINTS + EXECUTION + VERIFICATION + OUTPUT

Component	Purpose
Role	Establish relevant expertise and responsibilities.
Context	Provide background, previous decisions and source material.
Objective	Define the exact task and intended outcome.
Requirements	State what must be accomplished.
Constraints	Set boundaries, restrictions and items to preserve.
Execution	Describe the expected approach when the process matters.
Verification	Explain how correctness and completeness must be checked.
Output	Specify format, length, audience and presentation.

Prompt for outcomes, not unnecessary micromanagement

• Define the desired outcome, important constraints and quality standard.

• Allow the AI to choose a suitable method when the process itself is not important.

• Give procedural steps only when they materially affect correctness, safety or delivery.

• Not every simple request needs all eight formula components.

Priority order

Put the critical objective and mandatory requirements first. Include essential context and restrictions next, then supporting details and output requirements. For long source-based prompts, provide the source material and background before the final task instruction.

2. Role Definition and Context Engineering

Choose one relevant role

A role establishes the perspective and expertise expected. Use one primary role and add supporting expertise only when needed; stacking many titles does not automatically improve output.

Task	Example role
Research	Research analyst skilled in evidence evaluation and factual verification.
Programming	Software engineer skilled in architecture, debugging, security and testing.
Mathematics	Mathematics educator skilled in rigorous problem-solving.
Business	Business strategist skilled in market, financial and feasibility analysis.
Design	UI/UX designer skilled in usability, accessibility and visual hierarchy.

Define responsibilities and audience

• Ask the AI to analyze the problem, apply relevant expertise, evaluate alternatives, identify risks and verify important conclusions.

• Identify the intended audience: beginner, university student, developer, technical expert, business owner, general reader or evaluator.

• Audience determines terminology, complexity, explanation depth and tone.

Supply relevant context

• Project background, current status and previous decisions.

• Existing work, files, screenshots, datasets and authoritative specifications.

• Important preferences, resources, technical environment and limitations.

• Intended audience, deadline and desired outcome.

Establish a source of truth

Tell the AI which supplied document or file is authoritative. Require it to preserve source terminology, avoid inventing missing details, identify contradictions and disclose when the material is insufficient. Clearly distinguish source-supported content from external research or general knowledge.

Structure context and prevent overload

Use headings such as PROJECT BACKGROUND, CURRENT STATUS, PREVIOUS DECISIONS, AVAILABLE RESOURCES, RESTRICTIONS and EXPECTED RESULT. Include only details that affect the objective, implementation, constraints, correctness or output. Remove unrelated and duplicate information.

3. Define Objectives, Scope and Instructions

Make the task precise

Replace vague requests such as “explain AI” or “make it amazing” with an explicit task, audience, topics to cover, level of detail and expected format. Define subjective words such as “professional”, “advanced” or “innovative” in observable terms.

Define successful completion

• A functioning application with specified features and tests.

• A research report with credible evidence and citations.

• Study notes aligned with a supplied syllabus.

• A presentation with an exact slide count and coherent narrative.

• A calculation checked for method, units and precision.

• A comparison with stated criteria and a justified recommendation.

Separate tasks and set boundaries

For multi-part work, list each task separately (for example: research, analyze, compare, recommend and present). Define what is included and excluded, the expected depth, whether external research is allowed, and whether the AI should execute actions or only explain them. Do not expand scope without permission.

Prioritize competing objectives

A useful default priority is: accuracy, mandatory compliance, completeness, practical usefulness, efficiency, creativity and presentation. Change this order when the task requires a different priority.

Use direct, single-purpose instructions

• Prefer clear verbs: analyze, compare, calculate, verify, explain, implement, summarize, evaluate, recommend and generate.

• Separate content, formatting, technical, verification and delivery requirements.

• Use measurable quantities: exact item counts, lengths, dimensions, file types, languages or deadlines.

• Pair restrictions with the desired alternative behaviour.

Label instruction strength

Label	Meaning
MUST / MUST NOT	Non-negotiable requirement or prohibited action.
SHOULD	Strong preference; deviate only with a reason.
MAY	Optional improvement.

Check for conflicts such as “short but exhaustive” or “use only this file and include current external research.” If a conflict materially changes the result, identify it and ask for clarification rather than silently choosing.

4. Reasoning and Problem-Solving

Encourage careful analysis

For substantial tasks, ask the AI to understand the full problem, identify variables and dependencies, break complexity into parts, consider viable alternatives, evaluate trade-offs, examine failure cases and verify the result.

Decompose complex work

1. Understand the problem and intended outcome.

2. Identify requirements, constraints and dependencies.

3. Research or analyze the available information.

4. Develop suitable approaches and compare them.

5. Select the approach that best satisfies the requirements.

6. Execute the work in manageable stages.

7. Test or verify the result and correct identified issues.

8. Deliver the result with relevant limitations.

Do not impose this full process on simple questions. Use only the stages that add value.

Challenge assumptions and consider alternatives

• Ask what could be wrong, what assumptions are being made and whether another interpretation exists.

• Check for missing information, exceptions and conclusions that do not follow from evidence.

• For difficult decisions, compare practical options on accuracy, complexity, cost, efficiency, risk and suitability.

• Use normal, boundary, invalid and special test cases where relevant.

Request useful explanations, not private reasoning transcripts

Ask for the conclusion, important assumptions, key calculations, relevant evidence, concise justification and verification status. Do not demand hidden internal chain-of-thought; a concise, auditable explanation is more useful.

> The AI should distinguish facts, assumptions, inferences, opinions and recommendations. Never present an assumption as a confirmed fact.

5. Compulsory Double-Verification Protocol

This is a mandatory instruction for every important prompt. Every substantial response must undergo two separate review passes before submission.

First review — correctness and completeness

• Was the task interpreted correctly and is every requirement addressed?

• Are facts, sources, calculations and logic accurate?

• Are assumptions reasonable and clearly identified?

• Are any important components missing?

• Have all mandatory constraints been followed?

Second review — independent error detection

• Re-examine the complete result critically and independently.

• Look for overlooked details, contradictions, unsupported claims and hidden mistakes.

• Recheck calculations, formatting, completeness and whether the output meets the original objective.

• Correct every identified issue before finalizing.

Compulsory instruction to include in important prompts

COMPULSORY DOUBLE-VERIFICATION: Before submitting your final response, perform two separate review passes. FIRST REVIEW: verify accuracy, relevance, logic, completeness, calculations, assumptions and compliance against the original requirements. SECOND REVIEW: independently re-examine the entire result for overlooked requirements, factual or logical errors, incorrect calculations, missing information, unsupported claims, formatting problems and avoidable mistakes. Correct every identified issue. Cross-check important information using reliable sources, independent calculations, available tools or actual execution wherever appropriate. Never invent information, sources, test results or verification claims. If something cannot be verified, disclose that limitation. Do not declare completion until the result has been checked against the original objective and every mandatory requirement.

Match verification to the task

Task	Appropriate check
Mathematics	Recalculate independently; check method and units.
Research	Cross-check claims against credible, preferably primary sources.
Programming	Run relevant tests and inspect actual behaviour.
Data analysis	Validate input data, calculations and interpretations.
Writing / slides	Check clarity, grammar, facts, layout and consistency.
Business / academic	Check feasibility, assumptions, evidence and source or syllabus alignment.

6. Accuracy, Research and Hallucination Prevention

Prioritize factual accuracy

• Use reliable sources and prefer primary sources appropriate to the subject.

• Check publication dates and whether information is current.

• Verify important figures, technical details and quotations.

• Identify conflicting evidence and distinguish fact from interpretation.

• Do not fabricate facts, sources, citations, statistics or references.

General source hierarchy

1. Official government and regulatory sources.

2. Official technical documentation.

3. Original research papers and primary studies.

4. Reputable academic institutions.

5. Established industry publications.

6. Credible news organizations and expert commentary.

7. Community discussions and informal sources.

Choose sources appropriate to the topic. Official software documentation is usually best for API specifications; original studies are more suitable for scientific findings.

Citations and current information

For research-intensive work, request source title, publishing organization, date where available, direct link, supporting evidence and limitations. Citations must support the specific claims beside them. For current software versions, policies, admissions, prices, regulations or recent developments, require current research rather than relying only on stored model knowledge.

Handle disagreement transparently

1. Identify the disagreement and compare source authority and dates.

2. Examine the original evidence where possible.

3. Explain the most credible interpretation and disclose unresolved uncertainty.

Separate source and model contributions

When working from supplied documents, label what is directly supported by the material, what comes from external research or general background knowledge, and what is inferred or recommended. Never silently replace the source's framing or fill gaps with unsupported assumptions.

7. Creativity, Efficiency and Tool Use

Make creativity useful

• Explore overlooked opportunities, simpler approaches and practical alternatives.

• Consider user experience, automation and future scalability when relevant.

• Respect what may change and what must remain unchanged.

• Evaluate suggested improvements for feasibility, cost, complexity and value.

• Do not add unrelated, infeasible, expensive or unnecessarily complicated features.

Keep prompts concise but complete

Keep essential context, mandatory requirements, relevant constraints, verification instructions and expected output. Remove repetition, unrelated background, duplicate examples and excessive role descriptions. A longer prompt is not inherently better.

Specify response depth

Depth	Expected response
Short	Direct answer with essential information only.
Medium	Explanation with relevant examples and important details.
Detailed	Major aspects, examples, limitations and recommendations.
Exhaustive	In-depth treatment of all relevant areas, with structured analysis.

Clarification policy

If minor information is missing, make a reasonable assumption, state it briefly and proceed. If missing information materially affects correctness, safety, cost or the outcome, ask one focused clarification question.

Use tools selectively

• Web research for current or externally verifiable information.

• Code execution for calculations, programming and tests.

• File analysis for supplied documents and datasets.

• Image analysis for visual references.

• Connected applications for explicitly authorized external tasks.

Use a tool only when it materially improves the result. Check tool errors and returned values, stop when sufficient evidence is obtained, change approach after repeated failures and disclose unavailable tools or incomplete operations. Balance quality with time, latency and cost.

8. Multimodal Prompting and Output Design

Images and image editing

Explain what the image represents, what to inspect or create, and which elements must remain unchanged. Specify composition, perspective, environment, style, lighting, colours, aspect ratio, resolution, required text and elements to exclude. For edits, clearly separate changeable areas from elements that must remain identical.

Documents and multiple sources

• Identify each file and the operation required: summarize, extract, compare, edit or analyze.

• Specify whether the whole document or selected sections matter.

• Preserve required terminology and request page references when useful.

• Assign a distinct role to each source (for example, specification, visual reference and numerical dataset).

Audio and video

• Audio: specify transcription, summary, analysis or generation; language; speaker labels; timestamps; exact-wording requirements; and how unclear sections should be marked.

• Video: specify relevant timestamps, events, whether audio and visuals should be combined, chronological ordering and required summary depth.

Choose the right output format

Specify paragraphs, bullets, numbered steps, tables, Markdown, JSON, code, reports, presentations, checklists or structured documents. Define section order, table columns, item count, word limit, language, tone, technical depth and citation format.

Presentation quality

• Organize information logically and use consistent terminology.

• Explain technical terms when necessary and highlight important conclusions.

• Avoid repetition and overcrowding.

• Match detail and format to the user's purpose: students may need explanations; developers need complete code and setup; evaluators need measurable evidence; managers need concise decisions.

9. Tool Use, Autonomy and Execution Boundaries

Define permitted actions

For tool-enabled or agentic tasks, specify whether the AI may read files, search the web, analyze datasets, execute code, modify project files, run tests, create documents or interact with connected applications.

Set approval boundaries

• Allow necessary, ordinary and non-destructive actions that are within the requested scope.

• Allow in-scope file inspection, code edits and testing when implementation is requested.

• Ask before destructive operations, irreversible actions or external writes.

• Ask before purchases, financial commitments or expanding the project beyond its scope.

Require honest tool reporting

• Distinguish actual tool output from assumptions and estimates.

• Check errors and verify important returned values.

• Do not claim an action succeeded unless the tool confirms it.

• Report partial completion, failed operations and remaining limitations.

Avoid unnecessary loops

Use the most relevant tool, avoid repeated searches without a reason, stop when sufficient evidence is available and change approach if a tool repeatedly fails. Never use tools merely to appear thorough.

Completion standard

A task is complete only when the result addresses the objective, satisfies mandatory requirements, respects constraints, is accurate and logically consistent, has undergone both review passes, has been verified to the extent possible and discloses remaining uncertainties.

10. Model-Specific Prompting Guidance

A. GPT-5.6 Luna

Recommended approach:

1. Define the intended outcome and provide relevant context.

2. State hard constraints, success criteria and what must be preserved.

3. Establish autonomy and approval boundaries.

4. Avoid repetitive or unnecessarily restrictive instructions.

5. Allow an appropriate approach within the requirements.

6. Use tools when they materially improve accuracy.

7. Validate results and report honestly.

Reasoning effort guidance from the source handbook:

Setting	Suggested use
none / low	Simple tasks and routine work.
medium	General-purpose balanced tasks.
high	Complex analysis and problem-solving.
xhigh / max	Demanding multi-step or quality-first work where extra latency and token use are acceptable.

The best setting depends on workload. The source also distinguishes reasoning modes from reasoning-effort settings. Do not micromanage every step: provide the goal, context, constraints, permissions and success criteria.

B. Gemini 3.8 Flash

Recommended approach:

1. Be direct and precise; use clear headings or XML-style tags.

2. Separate context from instructions and define ambiguous terms.

3. Put critical requirements near the beginning.

4. Specify expected response depth and use examples for unusual requirements.

5. Provide large context before the final task.

6. Reference images, documents, audio and video explicitly.

7. Request verification of important conclusions.

Thinking-level guidance from the source handbook: low for quick, straightforward tasks; medium for general tasks and substantial coding; high for difficult reasoning, mathematics and complex multi-step tasks. The source identifies medium as default and says minimal is unsupported.

API settings and compatibility can change. Check current official API documentation before relying on model IDs, parameter support, function-call formats, multimodal payload rules or conversation-history requirements. API implementation details do not belong in every ordinary chat prompt.

11. Specialized Prompting Checklists

Research

• Define objective, scope, time period and geographic coverage.

• Specify source hierarchy, key questions, comparison criteria, citation format and report structure.

• Require evidence gaps, conflicting sources and uncertainty to be identified.

Academic learning

• Specify subject, topic, current knowledge level and syllabus.

• Define learning objectives, depth, examples, terminology and exam relevance.

• Ask for progressive explanations and the required notes format.

Mathematics

• Provide the complete problem, given values, required result, units and conditions.

• Specify a required method and precision, if applicable.

• Require independent recalculation, unit checks and final-answer verification.

Business strategy

• Describe the idea, market, target customers, budget, resources and competition.

• State operational constraints, revenue model, risks and time horizon.

• Require realistic assumptions, cost analysis, feasibility and practical recommendations.

Data analysis

• Describe dataset, source, columns and analysis objective.

• Specify missing-data policy, statistical requirements and visualizations.

• Require data-quality, calculation and interpretation checks.

Writing

• Define purpose, audience, tone, length and main message.

• List required details, prohibited content and output format.

• Check factual consistency, clarity, grammar and naturalness.

Image generation

• Specify subject, purpose, composition, perspective, environment, style and lighting.

• Define colours, aspect ratio, required text and excluded elements.

• Separate essential visual details from areas where creative freedom is allowed.

Decision-making

• State the decision, available options, objectives and constraints.

• Define evaluation criteria, risks and time horizon.

• Request a comparison, trade-off analysis and justified recommendation.

12. Universal Master Prompt — Reusable Template

Customize the sections to the task. Do not blindly include irrelevant sections.

ROLE AND RESPONSIBILITY
Act as an expert in the domain relevant to this task. Apply appropriate domain knowledge, analytical thinking, accuracy, creativity and practical problem-solving. Deliver a reliable, relevant and usable result.

1. CONTEXT
[Provide relevant background, documents, files, references and previous decisions.] Treat specified authoritative material as the source of truth. Do not invent missing information.

2. OBJECTIVE
[Explain exactly what must be accomplished and describe the intended outcome.]

3. REQUIREMENTS
[List mandatory requirements.] Distinguish MUST, SHOULD and optional improvements.

4. CONSTRAINTS
[List technical, practical, financial, time-related and content restrictions.] Preserve specified elements and avoid unrelated changes or unnecessary complexity.

5. EXECUTION
Understand the task, identify dependencies and risks, break complex work into manageable parts, consider suitable alternatives and choose the approach that best meets the requirements. Use tools and evidence when they materially improve the result.

6. ACCURACY AND SOURCES
Prioritize correctness. Do not fabricate facts, sources, calculations or technical details. Verify current information when required. Distinguish facts, assumptions, inferences and recommendations.

7. CREATIVITY AND OPTIMIZATION
Suggest practical improvements where appropriate. Prioritize simplicity, effectiveness, efficiency and feasibility. Do not add unnecessary features.

8. OUTPUT
[Specify structure, format, length, language, tone, audience and presentation.] Make the result complete, clear and directly usable.

9. COMPULSORY DOUBLE-VERIFICATION
Perform two separate review passes. First check accuracy, relevance, completeness, logic, calculations, interpretation and compliance. Then independently re-examine for overlooked details, mistakes, unsupported claims, contradictions and formatting problems. Correct issues. Cross-check important facts and calculations using suitable independent methods, sources or actual tests. Never claim verification that did not occur; disclose what remains unverified.

10. COMPLETION
Finish only when the result addresses the objective, satisfies mandatory requirements, respects constraints, has undergone both reviews and is verified to the extent possible. Disclose remaining uncertainty.

13. Golden Rules and Final Quality Scorecard

40 golden rules — condensed into practice

Area	Rules
Understanding	1. Define the objective. 2. Provide relevant context. 3. Identify the audience. 4. Establish the source of truth. 5. Preserve prior decisions. 6. Separate facts and assumptions.
Instructions	7. Use direct language. 8. Structure complex prompts. 9. State mandatory requirements. 10. Separate requirements from preferences. 11. Avoid conflicts. 12. Remove repetition. 13. Define scope. 14. Specify output.
Problem-solving	15. Analyze carefully. 16. Decompose complex tasks. 17. Compare alternatives when useful. 18. Examine risks and edge cases. 19. Challenge assumptions. 20. Use evidence-based conclusions.
Accuracy	21. Prioritize accuracy. 22. Use reliable sources. 23. Check current information. 24. Verify calculations. 25. Cross-check independently. 26. Review twice. 27. Correct errors. 28. Never fabricate verification.
Efficiency	29. Be concise but complete. 30. Avoid unnecessary clarification. 31. Use tools selectively. 32. Do not overcomplicate. 33. Encourage practical creativity. 34. Optimize for actual requirements.
Delivery	35. Match the user's purpose. 36. Format clearly. 37. State limitations. 38. Provide actionable results. 39. Check every mandatory requirement. 40. Do not declare completion without verification.

Final prompt quality scorecard

Criterion	Check
Clarity and context	Is the task unambiguous and is sufficient information provided?
Relevance and completeness	Does the result address the objective and every mandatory requirement?
Accuracy and reasoning	Are facts, calculations and conclusions correct and supported?
Creativity and efficiency	Are improvements useful, feasible and appropriately simple?
Reliability and verification	Are assumptions and uncertainties clear, and has the result been checked twice?
Presentation	Is the final output clear, consistent and usable?

Suggested rating: 9–10 excellent; 7–8 good; 5–6 acceptable but needs improvement; 3–4 poor; 0–2 unacceptable. A high average cannot excuse a critical failure such as fabricated citations, an incorrect core calculation, a violated mandatory restriction or a false claim of successful execution.

FINAL PRINCIPLE — UNDERSTAND → ANALYZE → REASON → CREATE → OPTIMIZE → VERIFY TWICE → CORRECT → DELIVER.
The goal is not to make AI produce more content or use more reasoning. It is to produce the most accurate, relevant, efficient, creative and dependable result for the actual task. Never submit important work without two reviews, appropriate cross-checks, correction of identified mistakes and honest disclosure of what remains unverified.